

SKILL WEEK 7

Task 1:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;
    int status;
    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }
    if (pid == 0) {
        // Child process
        printf("Child process started.\n");
        printf("Child PID: %d\n", getpid());
        sleep(3);
        printf("Child process completed.\n");
        exit(0);
    } else {
        // Parent process
        printf("Parent process waiting for child...\n");
        waitpid(pid, &status, 0);
        if (WIFEXITED(status)) {
            printf("Child exited normally.\n");
            printf("Child exit status: %d\n", WEXITSTATUS(status));
        }
        printf("Parent process completed.\n");
    }
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano s7.1.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc s7.1.c -o s7.1
ashishreddy@Ashishs-MacBook-Air ~ % ./s7.1
Parent process waiting for child...
Child process started.
Child PID: 1243
Child process completed.
Child exited normally.
Child exit status: 0
Parent process completed.
```

Task 2:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main() {
    char *path = getenv("PATH");

    if (path == NULL) {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH:\n%s\n\n", path);

    char *copy = strdup(path);
    char *directory = strtok(copy, ":");

    char filename[100];

    printf("Enter executable name to search: ");
    scanf("%99s", filename);

    while (directory != NULL) {
        char fullpath[500];

        snprintf(fullpath, sizeof(fullpath), "%s/%s", directory, filename);

        if (access(fullpath, X_OK) == 0) {
            printf("Executable found: %s\n", fullpath);
            free(copy);
            return 0;
        }
        directory = strtok(NULL, ":");
    }
}
```

```
    printf("Executable not found in PATH.\n");  
    free(copy);  
    return 0;  
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano s7.2.c  
ashishreddy@Ashishs-MacBook-Air ~ % gcc s7.2.c -o s7.2  
ashishreddy@Ashishs-MacBook-Air ~ % ./s7.2  
PATH:  
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/ws  
l/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64_  
_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c  
/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Users/venka/AppData/Local/Microsoft/WindowsApps:/snap/b  
in  
  
Enter executable name to search: ls  
Executable found: /usr/bin/ls
```